Katerina Smyrli

Robotics Engineer

katerinasmyrli@gmail.com   Athens, Greece   AlbertEinstein   example   example

Summary

Researching bipedal locomotion in robots and humans

Education

National Technical University of Athens, Control Systems Lab, Department of Mechanical Engineering — Athens, GR, 2018 – 2025

Imperial College London, Tribology Lab, Department of Mechanical Engineering — London, UK, 2017 – 2018
- Awarded with Distinction.

National Technical University of Athens, Control Systems Lab, Department of Mechanical Engineering — Athens, GR, 2012 – 2017
- Awarded with Distinction.

Experience

Athena Research Center, Adjunct Researcher — Marousi, GR, 2023 – present, 3 years
- Robotics Institute
- HERON
  - The Hellenic Robotics Center of Excellence

Control Systems Lab, National Technical University of Athens, Robotics Researcher — Athens, GR, 2018 – present, 8 years

Researcher in multiple EU-funded as well as private research projects, focusing on Dynamics and Control in space robotics.
- CoDyCO
  - Study of the Contact Dynamics during Object Capture on Orbit
- Study of a Waste Bin Weighing System
  - Analysis and simulation of Waste Collector System
- PRINCE
  - Passive Mechanical and Rendezvous Interface for Capture after End-of-Life
- DiGi-CHE Asia
  - Enhancing Digital Capacities in Higher Education for Asian Universities

TWI Hellas, Technical Project Manager — Athens, GR, 2023 – 2025, 2 years

Technical coordinator of Horizon Europe robotics projects.
- PALPABLE
  - Multi-sensing tool for Minimally Invasive Surgery
- SWAG
  - Soft Wearable Assistive Garments for Human Empowerment

University of Duisburg-Essen, Visiting Researcher — Duisburg, DE, 2018 – 2019, 1 year

Part of a PhD exchange with a focus on clinical gait analysis (German Academic Exchange Service - DAAD).
- Study of Gait Analysis Systems

Publications

A Virtual Gravity Controller for Efficient Underactuated Biped Robots 2025
Despoina Maligianni, Fotios Valouxis, Antonios Kantounias, Aikaterini Smyrli, Evangelos Papadopoulos
ieeexplore.ieee.org/document/11128793

Low-Cost Markerless Gait Analysis Using a Minimal Human Model 2025
Konstantina Tsintzira, Aikaterini Smyrli, Athanasios Mastrogeorgiou, Evangelos Papadopoulos
ieeexplore.ieee.org/document/11273621

User-Centered Design Approach for Development of an Assistive Soft Exosuit and Initial Results on User Requirements 2024
Gerdienke B. Prange-Lasonder, Kira Oberschmidt, Ilaria Pacifico, Marko Jamsek, Jan Babic, Federico Masiero, Lorenzo Masia, Aikaterini Smyrli, Erik C. Prinsen
link.springer.com/chapter/10.1007/978-3-031-77588-8_13

Guidelines For Optimal Human Mesh Generation Using Deep Learning-Driven Avatar Reconstruction For Gait Analysis 2024
Odysseas Stavrakakis, Athanasios Mastrogeorgiou, Aikaterini Smyrli, Evangelos Papadopoulos
ieeexplore.ieee.org/document/10769832

Gait Analysis with Trinocular Computer Vision Using Deep Learning 2023
Odysseas Stavrakakis, Athanasios Mastrogeorgiou, Aikaterini Smyrli, Evangelos Papadopoulos
ieeexplore.ieee.org/document/10328370

Modeling, Validation, and Design Investigation of a Passive Biped Walker with Knees and Biomimetic Feet 2022
Aikaterini Smyrli, Evangelos Papadopoulos
ieeexplore.ieee.org/document/9812129

Improved biped walking performance around the kinematic singularities of biomimetic four-bar knees 2022
Aikaterini Smyrli, Evangelos Papadopoulos
ieeexplore.ieee.org/document/9982027

Development of a passive biped robot digital twin using analysis, experiments, and a multibody simulation environment 2021
Christos Vasileiou, Aikaterini Smyrli, Anargyros Drogosis, Evangelos Papadopoulos
www.sciencedirect.com/science/article/abs/pii/S0094114X2100104X

A Generalized Model for Compliant Passive Bipedal Walking - Sensitivity Analysis and Implications on Bionic Leg Design 2021
Aikaterini Smyrli, Georgios A. Bertos, Evangelos Papadopoulos
asmedigitalcollection.asme.org/biomechanical/article-abstract/143/10/101008/1109461/A-Generalized-Model-for-Compliant-Passive-Bipedal

A methodology for the incorporation of arbitrarily-shaped feet in passive bipedal walking dynamics 2020
Aikaterini Smyrli, Evangelos Papadopoulos
ieeexplore.ieee.org/document/9196617

On the effect of semielliptical foot shape on the energetic efficiency of passive bipedal gait 2019
Aikaterini Smyrli, Mehdi Ghiassi, Andres Kecskemethy, Evangelos Papadopoulos
ieeexplore.ieee.org/document/8967565

Application of OpenPose deep learning algorithm for gait parameter identification 2019
Athanasios Mastrogeorgiou, Aikaterini Smyrli, Evangelos Papadopoulos, Andres Kecskemethy, Silke Gegenbauer
nereus.mech.ntua.gr/pdf_ps/eccomas19.pdf

Efficient Stabilization of Zero-Slope Walking for Bipedal Robots Following Their Passive Fixed-Point Trajectories 2018
Aikaterini Smyrli, Georgios A. Bertos, Evangelos Papadopoulos
ieeexplore.ieee.org/document/8460845

Theses

Design and control of anthropomorphic and energy-efficient robotic systems with applications in lower-limb prostheses. 2025
The thesis investigates bipedal locomotion through modeling, simulation, and control of bio-inspired walkers for applications in robotics and lower-limb prosthetics. It develops biomimetic foot and knee designs, informed by human gait data, and an energy-efficient control strategy to create robotic and prosthetic systems that effortlessly generate natural walking patterns.
Aikaterini Smyrli
phdtheses.ekt.gr/eadd/handle/10442/60169?locale=en

Robotic Grip - Design, development and evaluation of a tactile sensing device. 2018
This thesis investigates the design and development of a finger-like tactile sensing device for robotic manipulators to detect contact and predict slip. Integrating thermal and force sensors inspired by the human tactile system, the device is experimentally evaluated for contact detection and slip sensing across different forces and materials.
Aikaterini Smyrli
spiral.imperial.ac.uk/entities/publication/fa9e6e52-efb4-44c9-912f-6cd14535f826

Design, modeling and control of a bipedal walking robot with applications in human walking dynamics. 2017
This thesis investigates the passive dynamics of a bipedal walking model with compliant legs, semicircular feet, and a hip mass to better replicate human gait, particularly the double-stance phase and heelstrike dynamics.
Aikaterini Smyrli
dspace.lib.ntua.gr/xmlui/handle/123456789/46119?locale-attribute=en

Skills

Robotics

Workshop Organizer

Tailored to Move - Wearable Robotics for Motion Assistance 2026

Advanced tissue examination for diagnostics in surgical soft robotics 2025

Markerless Motion Analysis 2023

Awards

C. Mavroidis Award 2022, 2020
For excellence in Robotics and Automation.
C. Mavroidis

D. Thomaidis Award 2018-2022
For the publication of several research works.
National Technical University of Athens

Thomaidis Foundation Award 2017
For excellent academic performance.
Thomaidis Foundation

C. Papakyriakopoulos Award 2013
For excellence in Mathematics.
National Technical University of Athens

Scholarship from the Association of Greeks from Egypt 2012-2016
For excellent academic performance.
Association of Greeks from Egypt

Languages

Greek
Native speaker

English
Proficient (C2)

Italian
Basic (B2)

Interests

Crafts

Activities

Projects

Quantum Computing Jan 2018 – Jan 2018
Quantum computing is the use of quantum-mechanical phenomena such as superposition and entanglement to perform computation. Computers that perform quantum computations are known as quantum computers.
- Quantum Teleportation
- Quantum Cryptography

References

Professor John Doe

Professor Jane Smith